

Heaven's Light Is Our Guide
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
1st Year EVEN Semester Examination 2024
COURSE NO: Math 1213 **COURSE TITLE: Co-ordinate Geometry and Ordinary**
Differential Equation

FULL MARKS: 60

TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
(ii) Figures in the right margin indicate full marks.
(iii) Use separate answer script for each section.

<u>SECTION : A</u>		Marks	CO
Q.1. (a)	Determine the angle through which the axes must be rotated to remove the xy term in the equation $7x^2 - 6\sqrt{3}xy + 13y^2 = 16$. Also find the transformed equation. 30	5	1
(b)	Find the value of μ so that the equation $\mu x^2 + 4xy + y^2 - 4x - 2y - 3 = 0$ represents a pair of straight line. Also find the angle between them. 4	5	1
Q.2. (a)	Reduce the equation $8x^2 + 4xy + 5y^2 - 24x - 24y = 0$ to the standard form. Then find the vertex and focus.	5	1
(b)	Find the angle between two lines represented by the second degree homogeneous equation.	5	1
Q.3. (a)	Find the relations among the direction cosines and direction ratios. Find the direction cosines of the lines joining two points $(2, -3, 4)$ and $(-1, 2, 3)$.	5	2
(b)	A line make angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube prove that, (i) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{4}{3}$ (ii) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma + \sin^2\delta = \frac{8}{3}$	5	2
Q.4. (a)	Find the equations of the planes through lines of intersection of the planes $2x + y + 2z + 3 = 0$ and $x + 2y - z = 0$ and distance 1 unit from the origin.	5	2
(b)	Find the length and equation of shortest distance between two straight lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-4}{4}$ and $\frac{x-2}{2} = \frac{y-4}{4} = \frac{z-5}{5}$. 4	5	2

SECTION : B

Q.5. (a)	Find the differential equation whose solution $y = Ae^{2x} + Be^{-2x}$ where A and B are arbitrary constants.	3	3
(b)	Solve the differential equation $(x + y)dy + (x - y)dx = 0$.	3	3
(c)	Solve the initial value problem that consists of the differential equation $\frac{dy}{dx} = x^3 - 2xy$ and the initial condition $y(1) = 1$.	4	3
Q.6. (a)	What are the integrating factors of $ydx - xdy = 0$? Hence find the solution.	5	3
(b)	A circuit consists of an inductance of 0.05 henry, a resistance of 20ohms, a condenser of capacitance 100 microfarads, and an emf of $E = 100$ volts. Find i and q , given the initial conditions $q = 0, i = 0$ when $t = 0$. Also sketch it.	5	4
Q.7. (a)	Reduce $\frac{dy}{dx} + y = y^2$ to the standard form. Hence solve it.	4	3
(b)	Find the particular solutions of (i) $\frac{d^2y}{dx^2} - y = e^{-x}$ and (ii) $\frac{d^2y}{dx^2} + y = \sin x$	6	3
Q.8. (a)	Orthogonality of Legendre polynomials, to prove that $\int_{-1}^1 P_m(x)P_n(x)dx = 0$ if $n \neq m$.	5	4
(b)	Find the solution of the simple pendulum equation. Hence find the period of oscillation.	5	4

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

1st Year Even Semester Examination 2023

COURSE NO: Math 1213

COURSE TITLE: Coordinate Geometry and Ordinary

Differential Equation

FULL MARKS: 60

TIME: 3 HOURS

- N.B.** (i) Answer any **SIX** questions taking any **THREE** from each section.
(ii) Figures in the right margin indicate full marks.
(iii) Use separate answer script for each section.
(iv) Course Outcomes (COs)

SECTION : A

		Marks	COs
Q.1. (a)	A game developer is creating a 2D platformer game where the main character navigates an elliptical terrain, originally defined by the equation of an ellipse $9x^2 + 16y^2 = 144$. To enhance game play, the developer decides to shift the origin to the point (2, -3) to reflect the starting position of the character. Additionally, the developer wants to rotate the terrain by 45° to provide a more dynamic visual perspective. Calculate the new equation of the ellipse after applying the specified shift of the origin and rotation of axes.	06	CO1
(b)	A drone navigator algorithm uses the path given by $3x^2 + 4xy + y^2 + 6x + 2y - 7 = 0$. Determine that the lines are perpendicular to each other.	04	CO1
Q.2. (a)	Deduce $8x^2 + 4xy + 5y^2 - 24x + 2y = 0$ to the standard form.	05	CO1
(b)	Transform $11x^2 - 4xy + 14y^2 - 58x - 44y = 0$ to rectangular coordinates removing the term x , y and xy .	05	CO1
Q.3. (a)	Find the angles between two diagonals of a cube.	05	CO2
(b)	In robotics, a robotic arm moves along the line $\frac{x-1}{1} = \frac{y+2}{-1} = \frac{z-3}{2}$. A sensor is located at the point (4, -1, 6). Calculate the perpendicular distance from the sensor to the line of motion of the robotic arm.	05	CO2
Q.4. (a)	Find the equation of the line through the origin parallel to the line $x - y + 2z = 0 = 3x + y + z$.	05	CO2
(b)	Two airplanes are flying in space along the lines defined by the equation: Airplane 1: $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-4}{1}$, Airplane 2: $\frac{x-3}{1} = \frac{y-2}{2} = \frac{z+5}{-1}$. Determine whether these two airplanes will collide by finding the shortest distance between their flight paths.	05	CO2

SECTION : B

Q.5. (a)	In a network protocol design, the signal delay is modeled by a second-order differential equation $\frac{d^2s}{dt^2} - 4\frac{ds}{dt} + 3s = 0$. Solve the equation for the general solution.	06	CO3
(b)	If the initial conditions are $s(0) = 2$ and $\frac{ds}{dt}(0) = -1$, find the particular solution.	04	CO3
Q.6. (a)	Find the integrating factor of $\frac{dy}{dx} + \rho(x)y = q(x)$, hence solve $\frac{dy}{dx} + y = e^{-x}$, $y(0) = 0$, and plot it.	05	CO3
(b)	Solve $\frac{dy}{dx} + y = -y^2$, $y(0) = 1$.	05	CO3
Q.7.	Solve $\frac{dy}{dx} - iwy = e^{iyx}$ (i) when $\gamma \neq w$ and (ii) $\gamma = w$, hence solve $\frac{d^2y}{dx^2} + \omega^2y = \cos\omega x$.	10	CO3
Q.8. (a)	Write a short note on ordinary and singular points of a second order differential equation.	03	CO4
(b)	Solve in series the differential equation $\frac{d^2y}{dx^2} + \frac{1}{x}\frac{dy}{dx} - \frac{1}{x^2}y = 0$.	07	CO4

Heaven's Light Is Our Guide
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

1st Year Even Semester Examination 2022

COURSE NO: Math 1213 COURSE TITLE: Coordinate Geometry and Ordinary Differential Equation
 FULL MARKS: 60 TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
 (ii) Figures in the right margin indicate full marks.
 (iii) Use separate answer script for each section
 (iv) COs-Course Outcomes.

SECTION A

- | | COs | Marks |
|--|-----|-------|
| Q.1. (a) Define order and degree of a differential equation with example. | CO3 | 02 |
| (b) Form a differential equation from the relation
$y = ae^{3x} + be^{-4x} + ce^{5x}$. | CO3 | 04 |
| (c) Solve the initial value problem
$(2y \sin x \cos x + y^2 \sin x)dx + (\sin^2 x - 2y \cos x)dy = 0, y(0) = 3$. | CO3 | 04 |
| Q.2. (a) Define linear and Bernoulli differential equation with example. | CO3 | 02 |
| (b) Solve the initial value problem
$(2x - 5y)dx + (4x - y)dy = 0, y(1) = 4$. | CO3 | 04 |
| (c) Solve $(D^2 - 2D + 1)y = 2xe^x + 6e^x$. | CO4 | 04 |
| Q.3. (a) Solve the initial value problem
$\frac{d^2 y}{dx^2} + y = 3x^2 - 4 \sin x, y(0) = 0, y'(0) = 1$. | CO1 | 05 |
| (b) A coil having a resistance of 15 ohms and an inductance of 10 henries is connected to 90 volts supply. Determine the value of current after 2 seconds. ($e^{-3} = 0.05$) | CO3 | 05 |
| Q.4. (a) Find a series solution of the differential equation $2xy'' + y' + 2y = 0$ about $x = 0$. | CO4 | 05 |
| (b) A body weighing 8 lb falls from the rest toward the earth from a great height. As it falls air resistance act upon it and it is assumed that this resistance (in pounds) is numerically equal to $2v$, where v is the velocity (in feet per second). Find the velocity and distance fallen at time t seconds. | CO1 | 05 |

SECTION B

- | | | |
|--|-----|----|
| Q.5. (a) Transform the equation $9x^2 + 24xy + 2y^2 - 6x + 20y + 41 = 0$ in rectangular coordinates so as to remove the terms in x, y and xy . | CO1 | 05 |
| (b) Transform the equation $11x^2 - 4xy + 14y^2 - 58x - 44y + 126 = 0$ to new axes of x and y whose equations are the mutually perpendicular to the lines $x - 2y + 1 = 0$ and $2x + y - 8 = 0$ respectively. | CO2 | 05 |
| Q.6. (a) Reduce the equation $8x^2 + 4xy + 5y^2 - 24x + 24y = 0$ to the standard form and identify the conic. Find the equation of axes and eccentricity of the conic. | CO1 | 05 |
| (b) If transformation equation of $ax^2 + 2hxy + by^2 = 0$ is $a_1x^2 + 2h_1xy + b_1y^2 = 0$ when the direction at axes is turned through an angle θ , without change of origin, then show that $a + b$ and $ab - h^2$ are invariant. | CO1 | 05 |
| Q.7. (a) Find the direction cosines of the line which is perpendicular to the lines with direction cosines proportional to $(4, 3, 1)$ and $(2, -4, -5)$. | CO2 | 05 |
| (b) Find the equation of the plane perpendicular to the planes $x - 4y + z = 0$ and $3x + 4y + z = 2$ and at a unit distance from the origin. | CO2 | 05 |
| Q.8. (a) Find the equation of plane through the line $\frac{x-2}{2} = \frac{y+1}{2} = \frac{z}{1}$ which is parallel to the line $\frac{x}{1} = \frac{y}{3} = \frac{z}{-1}$. | CO2 | 05 |
| (b) Find the length and the equation of the shortest distance between the lines whose equations are $x + y = 0, z = 4$ and $\frac{x-1}{4} = \frac{y-2}{3} = \frac{z-36}{-6}$. | CO2 | 05 |

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

1st Year Even Semester Examination 2021

COURSE NO: Math 1213 COURSE TITLE: Co-ordinate Geometry and Ordinary Differential Equation

FULL MARKS: 60

TIME: 3 Hours

- N.B. (i) Answer any **SIX** questions taking any **THREE** from each section.
 (ii) Figures in the right margin indicate full marks.
 (iii) Use separate answer script for each section.

SECTION A

- | | | Marks |
|------|---|-------|
| Q.1. | (a) Transform the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to rectangular axes through the point $(2, -1)$ and inclined at an angle $\tan^{-1}(-4/3)$. [CO1] | 05 |
| | (b) Prove that the straight lines represented by the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will be equidistant from the origin, if $f^4 - g^4 = c(bf^2 - ag^2)$. [CO1] | 05 |
| Q.2. | (a) Show that the pair of lines joining the origin to the points of intersection of the curve $x^2/a^2 + y^2/b^2 = 1$ by the line $lx + my + n = 0$ are coincident if $a^2l^2 + b^2m^2 = n^2$. [CO1] | 05 |
| | (b) Find the equation of the circle inscribed in the triangle determined by the lines $2x - 3y + 21 = 0$, $3x - 2y - 6 = 0$, $2x + 3y + 9 = 0$. [CO1] | 05 |
| Q.3. | (a) Find the direction cosines of the line which is perpendicular to the lines with direction cosines proportional to $(1, -2, -2)$, $(0, 2, 1)$. [CO2] | 03 |
| | (b) Find the equations of the planes through the intersections of the planes $x - 2y + 3z + 4 = 0$ and $2x - 3y + 4z - 7 = 0$ and the point $(1, -1, 1)$. [CO2] | 03 |
| | (c) A line makes angles $\alpha, \beta, \gamma, \delta$ with four diagonals of a cube. Prove that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = 4/3$. [CO2] | 04 |
| Q.4. | (a) Find the equation of the plane which passes through the line $y/b + z/c = 1$, $x = 0$ and parallel to the line $x/a = z/c = 1$, $y = 0$. [CO2] | 05 |
| | (b) Find the shortest distance between the lines $(x - 3)/3 = (y - 8)/-1 = (z - 3)/1$, $(x + 3)/-3 = (y + 7)/2 = (z - 6)/4$. Find also equation and the points in which it meets the given lines. [CO2] | 05 |

SECTION B

- | | | |
|------|--|----|
| Q.5. | (a) Form the differential equation from the relation $y = ae^{3x} + be^{-3x} + cx^3$. [CO3] | 04 |
| | (b) Define exact differential equation and integrating function. [CO3] | 02 |
| | (c) Solve the initial value problem $(ye^x + 2e^x + y^2)dx + (e^x + 2xy)dy = 0$; $y(0) = 6$. [CO3] | 04 |
| Q.6. | (a) Solve $(x^2 - 3y^2)dx + 2xydy = 0$. [CO3] | 03 |
| | (b) A body weighting 8 lb falls from rest towards the earth from a great height. As it falls, air resistance act upon it, and we shall assume that this resistance (in pounds) is numerically equal to $2v$, where v is the velocity (in feet per second). Find the velocity and distance fallen at time t seconds. [CO3] | 03 |
| | (c) Solve $2 \frac{d^2y}{dx^2} - (dy/dx)^2 + 4 = 0$. [CO3] | 04 |
| Q.7. | (a) Solve the following initial value problem $d^2y/dx^2 - 4dy/dx + 13y = 8\sin 3x$, $y(0) = 1$, $y'(0) = 2$. [CO3] | 04 |
| | (b) Find the power series solution of the differential equation $d^2y/dx^2 + xdy/dx + (x^2 + 2)y = 0$. [CO4] | 06 |
| Q.8. | (a) Prove that $P_n(x) = \frac{1}{2^n} \cdot \frac{1}{n!} \frac{d^n}{dx^n} (x^2 - 1)^n$; where $P_n(x)$ has the usual meaning. [CO4] | 05 |
| | (b) Prove that $J_n(x)$ is the coefficient of t^n in the expansion of $e^{x/2}(t - 1/t)$. [CO4] | 05 |

Heaven's Light Is Our Guide
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
1st Year Even Semester Examination 2019

COURSE NO: Math 1213 COURSE TITLE: Co-Ordinate Geometry and Differential Equation
 FULL MARKS: 72 TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
 (ii) Figures in the right margin indicate full marks.
 (iii) Use separate answer script for each section.

SECTION : A

Marks

- Q.1. (a) What is transformation of co-ordinates? Prove that the value of $g_2 + f^2$ in the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ remains unaffected by orthogonal transformation without change of origin. 6
 (b) Transform the equation $17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$ to one in which there is no term in x , y and xy . 6
- Q.2. (a) Find the angle between the lines joining the origin to the point of intersection of $y = x + 1$, with $x^2 - 3y^2 + 2xy - 3x + 3y + 7 = 0$. 4
 (b) Find the lengths and equation of the axis of the conic $13x^2 - 18xy + 37y^2 + 2x + 14y - 2 = 0$. 8
- Q.3. (a) The direction cosines of two straight-lines are given by the relations $al + bm + cn = 0$ and $ul^2 + vm^2 + wn^2 = 0$. Find the conditions that the straight lines will be perpendicular and parallel. 6
 (b) Define direction cosine and direction ratio of a straight lines. Prove that if (l, m, n) are direction cosines of any line then, $l^2 + m^2 + n^2 = 1$. 6
- Q.4. (a) Find the equation of the plane which contains the line $\frac{x}{1} = \frac{y-3}{2} = \frac{z-5}{3}$, and which is perpendicular to the plane $2x + 7y + 3z = 1$ 6
 (b) Find the length and equations of the shortest distance (SD) between the two lines, $\frac{x+3}{-4} = \frac{y-5}{3} = \frac{z}{2}$ and $\frac{x+2}{-4} = \frac{y}{1} = \frac{z-7}{1}$ 6

SECTION : B

- Q.5. (a) A particle starting with velocity u moves in a straight line with a uniform acceleration f . Find the velocity and distance travelled in any time. 4
 (b) Find foci of the curve which satisfies the differential equation, $(1+y^2)dx - xdy = 0$ and pass through the point $(1, 0)$. 4
 (c) Show that $Ax^2 + By^2 = 1$, is the solution of $x \left\{ y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \right\} = y \frac{dy}{dx}$ 4
- Q.6. (a) Solve the initial value problem, $(2x-5y)dx + (4x-y)dy = 0$, $y(1) = 4$ 6
 (b) Suppose $n \neq 0$ or 1 , then show that the transformation $v = y^{1-n}$ reduces the Bernoulli equation $\frac{dy}{dx} + p(x)y = Q(x)y^n$ to a linear equation in v . Solve $\frac{dy}{dx} + y = xy$. 6
- Q.7. (a) Find the solution of the initial-value problem $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^x - 10\sin x$, $y(0) = 1$, $y'(1) = 2$. 6
 (b) Solve $(D^2 + 2D + 4)y = x^2 + e^x \sin 2x$. 6
- Q.8. (a) Solve $(2x+1)^2 \frac{d^2y}{dx^2} - 6(2x+1) \frac{dy}{dx} + 16y = 8(2x+1)^2$ 6
 (b) Find the differential equation for force free undamped motion. Then show that $x = C \cos \left(\sqrt{\frac{k}{m}} t + \phi \right)$ is the solution of this type of motion. Also find the period of the motion. 6

Heaven's Light Is Our Guide
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
1st Year Even Semester Examination 2018

COURSE NO: Math 1213 COURSE TITLE: Co-ordinate Geometry and Ordinary Differential Equation
 FULL MARKS: 70 TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
 (ii) Figures in the right margin indicate full marks.
 (iii) Use separate answer script for each section.

SECTION : A

- Q.1. (a) The direction of axes remaining the same, choose a new origin such that the new co-ordinate of the pair of points whose old co-ordinates are (5, -13) and (-3, 11) may be in the forms (h, k) and (-h, -k). 6
 (b) Transform the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to rectangular axes through the point (2, -1) inclined at an angle $\tan^{-1}(4/3)$ to the original axes. 6
- Q.2. (a) Prove that a homogenous equation of the 2nd degree represents two straight lines, real or imaginary which all pass through the origin. 6
 (b) Find the distance between the points of intersection of the straight line $x \cos \alpha + y \sin \alpha - P = 0$ with the straight lines $ax^2 + 2hxy + by^2 = 0$ 6
- Q.3. (a) If a line make angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube, then prove that $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = 4/3$ 4
 (b) Find the acute angle between the lines whose direction cosines are given by the relations $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$ 4
 (c) Find the equation of the plane through the points (2, 2, 1), (9, 3, 6) and perpendicular to the plane $2x + 6y + 6z = 9$ 4
- Q.4. (a) Find the distance of the point (1, 2, 3) from the plane $2x + 4y + 3z = 20$ measured parallel to the line $\frac{x}{2} = \frac{y}{-2} = \frac{z}{1}$ 6
 (b) Find the equation of the line of shortest distance (S.D.) and length of the S.D. between the lines $\frac{x-1}{4} = \frac{y-2}{3} = \frac{z-1}{-5}$ and $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-4}{-4}$ 6

SECTION : B

- Q.5. (a) Form the differential equation of $y = ae^{2x} + be^{-3x} + ce^x$ where a, b, c are parameters. 4
 (b) Solve $3e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0$ $(1 - e^x)^3 \tan y = \frac{1}{c}$ 4
 (c) Solve the initial value problem $(3x^2 + 9xy + 5y^2)dx - (6x^2 + 4xy)dy = 0, y(2) = -1$ 4
- Q.6. (a) Define exact differential equation. Solve the initial value problem $(3x^2y^2 - y^3 + 2x)dx + (2x^3y - 3xy^2 + 1)dy = 0, y(-2) = 1$ $x^3y - xy^3 + x^2 + y = 0 - 1$ 6
 (b) Suppose $n \neq 0$ or 1. Show that the transformation $v = y^{1-n}$ reduces the Bernoulli equation $\frac{dy}{dx} + p(x)y = Q(x)y^n$ to a linear equation in v. Solve the differential equation $\frac{dy}{dx} + y = xy^3$ 6
- Q.7. Solve the following differential equations
 (i) $(D^3 - 5D^2 + 8D - 4)y = e^{2x} + 2e^x + 3e^{-x}$ 4
 (ii) $(D^2 - 2D - 1)y = e^x \cos x$ 4
 (iii) $(D^2 - 4)y = x^2 e^{3x}$ 4

- Q.8. (a) A circuit has in series an electromotive force given by $E = 100 \sin(60t)$ V, a resistor of 2Ω , an inductor of $0.1H$ and a capacitor of $\frac{1}{260}$ farads. If the initial current and the initial charge on the capacitor are both zero, then find the charge on the capacitor at any time $t > 0$. 6
 (b) Solve the differential equations

(i) $(x+2)^2 \frac{d^2y}{dx^2} - (x+2) \frac{dy}{dx} + y = 3x + 4$

(ii) $(1+x^2) \frac{d^2y}{dx^2} + 1 + \left(\frac{dy}{dx}\right)^2 = 0$

RAJSHAH UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

1st Year Even Semester Examination 2017

COURSE NO: Math 1213 COURSE TITLE: Co-ordinate geometry & Ordinary Differential Equation

FULL MARKS: 72

TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
 (ii) Figures in the right margin indicate full marks.
 (iii) Use separate answer script for each section.

SECTION : A

- Q.1. (a) If by the rotation of the rectangular co-ordinate axes about the origin, the expression $ax^2 + 2hxy + by^2$ changes to $a'x'^2 + 2h'x'y' + b'y'^2$. Show that $a + b = a' + b'$ and $ab - h^2 = a'b' - h'^2$. 6
- (b) Determine the equation of parabola $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$ after rotating of axes through 45° . 6
- Q.2. (a) Prove that the straight lines represented by the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will be equidistance from the origin if $f^4 - g^4 = bf^2 - ag^2$. 6
- (b) Find the area of the rectangle formed by the lines $9x^2 - 6xy + y^2 + 45x - 15y + 54 = 0$ and $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$. 6
- Q.3. (a) If (l_1, m_1, n_1) and (l_2, m_2, n_2) be the direction cosines of any two lines AB and CD and θ be the angle between them then show that $\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$. 6
- (b) Find the equation of the plane through the points $(1, -2, 2)$, $(-3, 1, -2)$ and perpendicular to the plane $2x + y - z + 6 = 0$. 6
- Q.4. (a) Find the equation of the line perpendicular to both the lines $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z+2}{3}$ and $\frac{x+2}{2} = \frac{y-5}{-1} = \frac{z+3}{2}$ and passing through their intersection point. 7
- (b) Find the angle between the line $2x + 4y - 2z + 3 = 0 = 4x - 2y + 6z + 5$ and the plane $5x - 4y + 3z - 5 = 0$. 5

SECTION : B

- Q.5. (a) Define linear and non-linear differential equation with example. Find the DE of all circles passing through the origin. 5
- (b) Solve the ODE $x^2 dy + y(x+y) dx = 0$. 4
- (c) Solve the initial value problem $(2x \cos y + 3x^2 y) dx + (x^3 - x^2 \sin y - y) dy = 0$; $y(0) = 2$. 3
- Q.6. (a) What is integrating factor. Solve the ODE by first finding an integrating factor $(2xy^2 + y) dx + (2y^3 - x) dy = 0$. 6
- (b) Solve the initial value problem $(x^2 + 1) dy + (4xy - x) dx = 0$ and the initial condition is $y(2) = 1$. 6
- Q.7. (a) Solve $(D^2 - 5D + 6)y = e^{3x} + e^{2x} \cos x$. 5
- (b) Solve $y = px + p^3$ where $p = \frac{dy}{dx}$. 2
- (c) Solve $(D^2 - 4)y = x^2 e^{2x} + \sin 3x$. 5
- Q.8. (a) Find the general solution of $\frac{d^2 y}{dx^2} + y = \tan x \sec x$. 7
- (b) A circuit has in series an electromotive force given by $E = 100 \sin(40t)$ V, a resistor of 10Ω and an inductor of $0.5H$. If the initial current is zero, find the current at time $t > 0$. 5
